

Calculus of Variation
Assignment 8



Q1 Choose the extremal(s) of the functional :

$$F(y) = \int_0^1 (y^2 + y'^2 + 2ye^x) dx, \quad y(0) = 0, \quad y(1) = 1$$

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|---|---|
| a) $y = \left(\frac{1-e/2}{\cosh 1}\right) \cosh x + \frac{1}{2}xe^x$ | b) $y = \left(\frac{1+e/2}{\cosh 1}\right) \cosh x + \frac{1}{2}xe^x$ |
| c) $y = \left(\frac{1-e/2}{\sinh 1}\right) \sinh x + \frac{1}{2}xe^x$ | d) $y = \left(\frac{1+e}{\sinh 1}\right) \sinh x + \frac{1}{2}xe^x$ |
| e) $y = \left(\frac{1+e/2}{\sinh 1}\right) \sinh x + \frac{1}{2}xe^x$ | f) $y = \left(\frac{1-e}{\sinh 1}\right) \sinh x + \frac{1}{2}xe^x$ |

Q2 Choose the extremal(s) of the functional :

$$I = \int_{x_1}^{x_2} \left(x (y')^2 - yy' + y \right) dx$$

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|--|--|
| a) $y = -\frac{c_1}{2x^2} + c_2$ | b) $y = -\frac{c_1}{x^2} + c_2$ |
| c) $y = \frac{c_1}{2} \ln x + c_2 x$ | d) $y = \frac{c_1}{2} \ln x + \frac{c_2}{x}$ |
| e) $y = -\frac{c_1}{x^2} + c_2 x$ | f) $y = \frac{c_1}{2} \ln x + c_2$ |

Q3 Choose the extremal(s) of the functional :

$$J(y) = \int_0^1 \frac{y'^2}{x^3} dx$$

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|------------------------------|-------------------------------|
| a) $y = \frac{cx^3}{2} + b$ | b) $y = \frac{cx^4}{8} + b$ |
| c) $y = \frac{3cx^2}{2} + b$ | d) $y = \frac{cx^4}{8} + bx$ |
| e) $y = \frac{cx^2}{2} + bx$ | f) $y = \frac{3cx^4}{2} + bx$ |